GWP Estimation — 25 m² Studio Apartment

Tool: eLCA v1.0.1 (bauteileditor.de) · Database: OBD_2024_I_A2 · Scope: A1–A3 + C3 + C4 + Maintenance

---

Minimalistic scope applied as instructed. Only primary structural elements modelled — floor slab, exterior walls, flat roof. Windows, doors, MEP systems, furniture, and operational energy excluded. Results represent embodied carbon of the building shell only.

SECTION 01

Results Summary

PROJECT: NEOBIM_25 M² STUDIO · AS OF 20.05.2026 · 50-YEAR SERVICE LIFE

GWP total impact	5.731 kg CO_2-eq / m²NGF
GWP total — whole studio (25 m²)	143.3 kg CO_2-eq
PENRT (non-renewable primary energy)	57.65 MJ / m²NGF
PERT (renewable primary energy)	35.19 MJ / m²NGF

Scope included: `A1–A3` `C3` `C4` `Maintenance` *· B6 operational energy = 0 (excluded) · Module D excluded · All values per m²NGF over 50-year reference period.*

SECTION 02

GWP Breakdown by Lifecycle Stage

STAGE	DESCRIPTION	GWP (KG CO_2-EQ/M^2NGF)	SHARE (%)
A1–A3	Product stage — raw material extraction + manufacturing	2.259	39.4%
C3	Waste processing — end of life treatment	3.288	57.4%
C4	Disposal	0.071	1.2%
B2/B4	Maintenance / servicing (Instandhaltung)	0.113	2.0%
Total		**5.731**	**100%**

Why is C3 higher than A1–A3? The exterior wall uses a timber frame construction. Timber stores biogenic carbon during its service life. At end of life, when the wood is processed or incinerated, that stored carbon is released — this appears as a large positive GWP-biogenic value in C3 (3.07 kg CO_2-eq/m^2). This is technically correct per EN 15804+A2 biogenic carbon accounting.

SECTION 03

Building Elements Modelled

ELEMENT	DIN 276	AREA / QTY	CONSTRUCTION TYPE
Floor slab	350	25 m²	RC slab 180mm + floating screed + linoleum
Exterior walls	330	50 m²	Timber frame, OSB + mineral wool, U = 0.096 W/m²K
Flat roof	**361**	**25 m²**	**Flachdach / Stahlbeton (Vorlage template)**

Exterior wall area estimated as 4 sides × 5m width × 2.7m height ≈ 54 m², rounded to 50 m² for simplicity.

SECTION 04

Assumptions

- Net floor area (NGF): 25 m² — single-storey, single-room studio
- Building service life: 50 years (DIN EN 15978 default for residential)
- Tool: eLCA v1.0.1 — bauteileditor.de (BBSR, Germany)
- Database: ÖKOBAUDAT OBD_2024_I_A2
- Scope: A1–A3 + C3 + C4 + Maintenance. B6 operational energy = 0. Module D excluded.
- Only primary structural shell modelled: floor, walls, roof
- Windows, doors, MEP, furniture excluded — minimalistic scope as instructed
- Exterior wall assumed as timber frame — U = 0.096 W/m²K (typical for lightweight construction)
- Operational energy excluded — no heating system modelled (B6 = 0)
- Results represent embodied carbon of structural shell only; whole-building GWP would be 20–30% higher

SECTION 05

Interpretation

The estimated embodied GWP of the 25 m² studio shell is **143.3 kg CO_2-eq total**, equivalent to **5.73 kg CO_2-eq/m²NGF** over 50 years.

METRIC	THIS STUDY	TYPICAL BENCHMARK	NOTE
GWP / m²NGF (shell only)	5.73 kg CO_2-eq	200–400 kg CO_2-eq	Lower — shell only, no MEP, no fit-out

METRIC	THIS STUDY	TYPICAL BENCHMARK	NOTE
Total studio GWP	**143.3 kg CO_2-eq**	**5,000–10,000 kg CO_2-eq**	**Lower — structural shell only**

The low result relative to benchmarks is expected and intentional — minimalistic scope excludes the majority of building components. In a production BIM-LCA workflow, element quantities would be extracted automatically from the IFC model, eliminating manual area estimation and enabling full-building scope.